Xiaocan Li

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Research Interests

Magnetic reconnection

Particle acceleration mechanisms. Power-law formation. Particle transport. Plasma energization. Turbulence generated by 3D magnetic reconnection. Turbulent reconnection. Reconnection rate. Relativistic reconnection. Reconnection-driven solar jets.

Solar flares and solar eruptions

Large-scale particle acceleration and transport. Nonthermal emissions produced by the energetic particles. The origins of solar energetic particles. Magnetic reconnection, termination shocks, and plasma turbulence in solar flares.

Particle acceleration and transport

Transport theory and modeling. Shock acceleration. Turbulence acceleration. Stochastic processes.

Education

The University of Alabama in Huntsville — Huntsville, AL, USA

Ph.D. in Space Science — 2011 – 2016

Dissertation: "Numerical Simulation of Particle Acceleration during Solar Flares". Advisor: Gang Li

Peking University — Beijing, China

M.S. in Space Physics — 2008 – 2011

Tongji University — Shanghai, China

B.S. in Surveying and Mapping Engineering — 2004 – 2008

Research Experiences

Dartmouth College — Hanover, NH

Research Scientist — Feb. 2022 – present

I study particle and transport processes in solar physics and astrophysics using MHD + transport equation and particle-in-cell simulations, develop theories of particle acceleration processes, and compare the results with emission observations.

Dartmouth College — Hanover, NH

Research Associate B — Feb. 2020 – Jan. 2022

I studied magnetic reconnection and particle energization processes in solar physics and astrophysics. Specifically, I performed PIC simulations to investigate the reduction of the reconnection rate in high-β plasmas and particle acceleration in high-σ relativistic plasmas, and I used MHD + transport equation to study particle acceleration and transport in solar flares. Mentor: Yi-Hsin Liu

Los Alamos National Laboratory — Los Alamos, NM

Postdoctoral Research Associate — Feb. 2017 – Feb. 2020

I studied particle acceleration during magnetic reconnection in solar flares. Specifically, I explored the primary particle acceleration mechanisms using kinetic particle-in-cell simulations, developed particle acceleration models, solved the models coupled with MHD simulations, and compared the results with solar flare observations. Mentors: Fan Guo, Hui Li

The University of Alabama in Huntsville — Huntsville, AL

Postdoctoral Research Assistant — Sept. 2016 – Jan. 2017

I studied the interaction of plasma with electric tether using kinetic particle-in-cell simulations. Mentor: Gary Zank

Los Alamos National Laboratory — Los Alamos, NM

Visiting Student — May 2014 – Aug. 2016

I studied particle energization during magnetic reconnection using the particle-in-cell code VPIC. I focused on the roles of particle guiding-center drift motions, flow compression and shear, and pressure anisotropy in particle energization. Mentors: Fan Guo, Hui Li

The University of Alabama in Huntsville — Huntsville, AL

Graduate Research Assistant — May 2012 – Aug. 2016

My dissertation was on numerical simulations of particle acceleration during solar flares. I developed a test-particle code to track charged particles in chaotic and turbulent magnetic fields. I used the kinetic particle-in-cell code VPIC to study particle acceleration during magnetic reconnection. Advisor: Gang Li

Los Alamos National Laboratory/Center for Nonlinear Studies — Los Alamos, NM

Graduate Research Assistant — Jun. 2013 – Aug. 2013

I studied the magnetic field amplification in supernova remnant shocks using MHD simulations and particle acceleration in the shock downstream by solving the Parker transport equation coupled with the MHD fields. Mentors: Fan Guo, Hui Li

Peking University — Beijing, China

Graduate Research Assistant — Sept. 2009 – Jun. 2011

I studied electron transport using the Monte Carlo code Geant4. Advisor: Hongfei Chen

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Honors and Awards

Postdoctoral Distinguished Performance Award Honorable Mention, Los Alamos National Lab — 2019

NASA Earth and Space Science Fellowship (NESSF), NASA — 2013 – 2016

SPD Studentship Award, Solar Physics Division, AAS — 2015

DPP Student Travel Grant Award, Division of Plasma Physics, APS — 2014

Teaching and Mentoring Experiences

Teaching:

Graduate Teaching Assistant, Department of Physics, The University of Alabama in Huntsville — Spring 2012

Grading for PH421 (Thermal & Statistical Physics)

Graduate Teaching Assistant, Department of Physics, The University of Alabama in Huntsville — Fall 2011

Grading for PH113 (General Physics with Calculus III) and teaching PH114 (General Physics Labs)

Tutorial:

A tutorial on running reconnection simulations using VPIC — 2021

Material: https://github.com/xiaocanli/vpic_reconnection_tutorial

Reconnection Simulation using `Athena++` — 2022

Material: https://github.com/xiaocanli/athena_reconnection

A tutorial on using stochastic-parker — 2022

Material: https://stochastic-parker.readthedocs.io/en/latest/usage.html

Mentoring/Technical Support:

Shan-Chang Lin, Graduate@Dartmouth College — 2020 – present

Helping with simulations and assisting in studying reconnection rate and magnetotail reconnection

Sarah Peery, Graduate@Dartmouth College — 2020 – present

Helping with simulations and assisting in studying relativistic reconnection

Matthew Goodbred, Undergraduate@Dartmouth College — 2020 – 2022

Helping with simulations and assisting in studying energy conversion in magnetic reconnection

Giulia Murtas, Postdoc@LANL — 2022 – present

Assisting and collaborating in modeling particle acceleration and transport in large-scale magnetic reconnection

Joshua Goodwill (*Graduate@University of Delaware*), LANL Summer Intern — 2023 – present

Assisting in studying Kelvin-Helmholtz instability using kinetic simulations

Grant Johnson (*Graduate@Princeton University*), LANL Summer Intern — 2021 – 2022

Assisting in studying particle acceleration in magnetic reconnection

Senbei Du (*Graduate@The University of Alabama in Huntsville*), LANL Visiting Student — 2016 – 2019

Helping with simulations and assisting in studying plasma energization in magnetic reconnection

Dylan Ma (*Los Alamos High School*), LANL Intern — 2018 – 2019

Helping with debugging, running simulations, and analyzing simulation results

David Fink (*Undergraduate@Emory University*), UAH & MSFC REU 2012
Helping with analyzing observational data

Grants & Computing Allocations

Research Grants:

Co-I, *NASA Heliophysics Theory, Modeling and Simulations* (*HTMS*), *$150k* 2023 – 2026
"Particle Acceleration and Transport in Coronal Solar Flare Regions" (PI: Fan Guo)

Co-I, *NASA Heliophysics Supporting Research* (*HSR*), *$150k* 2023 – 2026
"Magnetic flux ropes and magnetic loops in Earth's plasma sheet: their properties and role in particle energization" (PI: Weijie Sun)

PI, *NASA Living With a Star Science* (*LWS*), *$647k* 2021 – 2025
"The Acceleration of Energetic Particles in Solar Flares and Their Transport in Solar Eruption Regions"

Institutional PI, *NSF Astronomy & Astrophysics Grants Program* (*AAG*), *$178k* 2021 – 2024
"Achieving a New Understanding of Solar Flare Termination Shocks" (PI: Fan Guo)

Co-I, *NASA Heliophysics Supporting Research* (*HSR*), *$167k* 2021 – 2024
"Exploring Time-Dependent Ionization in Solar Eruption Models Combining with Energetic Particle Acceleration" (PI: Chengcai Shen)

Institutional PI, *Purdue University–NASA prime, $23k* 2021 – 2022
"Radiation and Polarization Signatures From 3D Relativistic Turbulent Magnetic Reconnection In Blazars" (PI: Dimitrios Giannios)

Institutional PI, *LANL–NASA prime, $116k* 2020 – 2023
"Exploring Energy Release and Conversion in Solar Eruptive Events Using Multi-wavelength Observations and Numerical Simulations" (PI: Bin Chen)

Computing Allocations:

PI, NERSC's ERCAP, 7k CPU node-hours and 2.5k GPU node-hours/year 2023

PI, NERSC's ERCAP (Energy Research Computing Allocations Process), ∼25k node-hours/year 2022

Co-I, TACC's Large-Scale Community Partnerships (LSCP) project, 890k node-hours/year 2021 – 2023

PI, Institutional Computing Project 2019, Los Alamos National Lab, 10M core-hours 2019 – 2020

PI, Institutional Computing Project 2018, Los Alamos National Lab, 8.8M core-hours/year 2018 – 2020

Co-I, NERSC's High-Impact Science at Scale program 2018, 60M core-hours 2018 – 2019

Professional and Community Service

Review Activities:

Guest Editor, *Frontiers in Astronomy And Space Sciences* 2021 – present

Referee, *50 papers for 21 journals* 2016 – present
ApJ, ApJL, ApJS, PRL, PRX, MNRAS, JGR, GRL, PoP, Reviews of Modern Plasma Physics, Communications Physics, Frontier, Astrophysics and Space Sciences, Solar Physics, Plasma Science and Technology, Galaxies, Sensors, Atmosphere, Fluids, Entropy, Universe

Panelist, *2 NASA grant panel, 1 NSF grant panel* 2022, 2023

Mail-in Reviewer, *1 grant proposal submitted to Swiss National Science Foundation* (*SNSF*) 2023

Mail-in Reviewer, *1 application submitted for Secretary's Research Prizes of Smithsonian Institute* 2023

Mail-in Reviewer, *2 grant proposals submitted to NASA funding programs* 2019, 2020

Judge, *Outstanding Student Paper Award of the AGU Fall Meeting* 2017 – 2019, 2021 – 2022

Meeting Activities:

Session Chair, *AGU Fall Meeting at San Francisco, CA* 2023
Topic: "Magnetic Reconnection: Connecting Ideas from Observations, Experiments, Models, and Theory"

Session Convener, *AGU Fall Meeting at Chicago, IL* 2022
Topic: "Magnetic Reconnection: Connecting Ideas from Observations, Experiments, Models, and Theory"

Early Career Convener, *AGU Fall Meeting at Chicago, IL* 2022
Topic: "Energetic Particles from the Sun to the Heliosphere"

Session Convener, *AGU Fall Meeting at San Francisco, CA (19), Online (20)* 2019-2020
Topic: "Plasma Energization, Particle Acceleration, and High-Energy Emission in Solar Flares"

Organizer, *One-week workshop at Los Alamos, NM* 2019
Topic: "Magnetic Reconnection and Particle Acceleration in Solar Flares"

Session Chair, *AGU Fall Meeting at Washington, D.C.* 2018
Topic: "Plasma Energization, Particle Acceleration, and High-Energy Emission in Solar Flares"

Computer Skills

HPC experience at NERSC, LANL, and TACC 2013 – present

Languages: Python, Fortran, C, C++, and Julia

Parallelization: MPI, OpenMP, CUDA, and Kokkos

Building: Make and CMake

Debugging: GDB, Valgrind, and Arm DDT

Data analysis and visualization: Python libraries, ParaView, IDL

Others: HDF5, FFTW3

Numerical codes: Athena/Athena++ MHD code, VPIC code, WarpX PIC code, test-particle code (`https://git.io/fxQYy`), stochastic code to solve the Parker transport equation (`https://git.io/fxQY1`)

Journal Articles

1000+ citations in total according to Google Scholar

As First Author

12. **Xiaocan Li**, Fan Guo, Yi-Hsin Liu, and Hui Li. "A Model for Nonthermal Particle Acceleration in Relativistic Magnetic Reconnection." The Astrophysical Journal 954 L37 (2023).
11. **Xiaocan Li**, Fan Guo, Bin Chen, Chengcai Shen, and Lindsay Glesener. "Modeling Electron Acceleration and Transport in the Early Impulsive Phase of the 2017 September 10 Solar Flare." The Astrophysical Journal, 932:92 (11pp), 2022 June 20
10. **Xiaocan Li**, Fan Guo, and Yi-Hsin Liu. "The acceleration of charged particles and formation of power-law energy spectra in nonrelativistic magnetic reconnection." Physics of Plasmas 28, no. 5 (2021): 052905 (**Featured Article**).
9. **Xiaocan Li**, and Yi-Hsin Liu. "The Effect of Thermal Pressure on Collisionless Magnetic Reconnection Rate." The Astrophysical Journal 912, no. 2 (2021): 152.
8. **Xiaocan Li**, Fan Guo, Hui Li, Adam Stanier, and Patrick Kilian. "Formation of Power-law Electron Energy Spectra in Three-dimensional Low-β Magnetic Reconnection." The Astrophysical Journal 884, no. 2 (2019): 118.
7. **Xiaocan Li**, Fan Guo, Hui Li. "Particle acceleration in kinetic simulations of non-relativistic magnetic reconnection with different ion-electron mass ratio" The Astrophysical Journal, 879:5 (12pp), 2019 July 1.
6. **Xiaocan Li**, Fan Guo, Hui Li, Shengtai Li. "Large-scale Compression Acceleration during Magnetic Reconnection in a Low-β Plasma." The Astrophysical Journal 866, no. 1 (2018): 4.
5. **Xiaocan Li**, Fan Guo, Hui Li, Joachim Birn. "The roles of fluid compression and shear in electron energization during magnetic reconnection." The Astrophysical Journal, 855:80 (2018)
4. **Xiaocan Li**, Fan Guo, Hui Li, and Gang Li. "Particle Acceleration during Magnetic Reconnection in a Low-beta Plasma." The Astrophysical Journal 843.1 (2017): 21.
3. **Xiaocan Li**, Fan Guo, Hui Li, and Gang Li. "Nonthermally Dominated Electron Acceleration during Magnetic Reconnection in a Low-beta Plasma." The Astrophysical Journal Letters, 811:L24 (5pp), 2015 October 1
2. **Xiaocan Li**, Brahmananda Dasgupta, and Gang Li. "Energization of charged particle in a time-dependent chaotic magnetic field with an implication of the production of seed particles in solar energetic particle events." Advances in Space Research 53, no. 8 (2014): 1153-1161.

1. **Xiaocan Li**, Hongfei Chen, Yongqiang Hao, Hong Zou, and Weihong Shi. "Investigation of electrons inside the satellite by the Geant4 simulation." Science China Technological Sciences 54, no. 9 (2011): 2271-2275.

As Second Author

10. Fan Guo, **Xiaocan Li**, Omar French, Qile Zhang, William Daughton, Yi-Hsin Liu, William Matthaeus, Patrick Kilian, Grant Johnson, and Hui Li. "Comment on "Nonideal Fields Solve the Injection Problem in Relativistic Reconnection"." Physical Review Letters 130, no. 18 (2023): 189501.
9. Kevin J. Genestreti, **Xiaocan Li**, Yi-Hsin Liu, James L. Burch, Roy B. Torbert, Stephen A. Fuselier, Takuma Nakamura et al. "On the origin of" patchy" energy conversion in electron diffusion regions." Physics of Plasmas 29, 082107 (2022)
8. Haocheng Zhang, **Xiaocan Li**, Dimitrios Giannios, Fan Guo, Hannes Thiersen, Markus Böttcher, Tiffany Lewis, and Tonia Venters. "Radiation and Polarization Signatures from Magnetic Reconnection in Relativistic Jets. II. Connection with γ-Rays." The Astrophysical Journal 924, no. 2 (2022): 90.
7. Fan Guo, **Xiaocan Li**, William Daughton, Hui Li, Patrick Kilian, Yi-Hsin Liu, Qile Zhang, and Haocheng Zhang. "Magnetic Energy Release, Plasma Dynamics and Particle Acceleration during Relativistic Turbulent Magnetic Reconnection." The Astrophysical Journal 919, no. 2 (2021): 111.
6. Haocheng Zhang, **Xiaocan Li**, Dimitrios Giannios, Fan Guo. "First-principles Prediction of X-Ray Polarization from Magnetic Reconnection in High-frequency BL Lacertae Objects." The Astrophysical Journal 912, no. 2 (2021): 129.
5. Haocheng Zhang, **Xiaocan Li**, Dimitrios Giannios, Fan Guo, Yi-Hsin Liu, and Lingyi Dong. "Radiation and Polarization Signatures from Magnetic Reconnection in Relativistic Jets. I. A Systematic Study." The Astrophysical Journal 901, no. 2 (2020): 149.
4. Patrick Kilian, **Xiaocan Li**, Fan Guo, and Hui Li. "Exploring the acceleration mechanisms for particle injection and power-law formation during trans-relativistic magnetic reconnection." The Astrophysical Journal 899, no. 2 (2020): 151.
3. Fan Guo, **Xiaocan Li**, William Daughton, Hui Li, Yi-Hsin Liu, Wangcheng Yan, Dylan Ma, and Patrick Kilian. "Determining the Dominant Acceleration Mechanism during Relativistic Magnetic Reconnection in Large-scale Systems." The Astrophysical Journal Letters 879, no. 2 (2019): L23.
2. Haocheng Zhang, **Xiaocan Li**, Fan Guo, and Dimitrios Giannios. "Large-Amplitude Blazar Polarization Angle Swing as a Signature of Magnetic Reconnection." The Astrophysical Journal Letters 862, no. 2 (2018): L25.
1. Fan Guo, **Xiaocan Li**, Hui Li, William Daughton, Bing Zhang, Nicole Lloyd-Ronning, Yi-Hsin Liu, Haocheng Zhang, and Wei Deng. "Efficient production of high-energy nonthermal particles during magnetic reconnection in a magnetically dominated ion–electron plasma." The Astrophysical Journal Letters 818, no. 1 (2016): L9.

As Supporting Author

21. Haocheng Zhang, Alan Marscher, Fan Guo, Dimitrios Giannios, **Xiaocan Li**, and Michela Negro. "First-Principle-Integrated Study of Blazar Synchrotron Radiation and Polarization Signatures from Magnetic Turbulence." Accepted to The Astrophysical Journal (2023).
20. Xiangliang Kong, Bin Chen, Fan Guo, Chengcai Shen, **Xiaocan Li**, Jing Ye, Lulu Zhao et al. "Numerical Modeling of Energetic Electron Acceleration, Transport, and Emission in Solar Flares: Connecting Loop-top and Footpoint Hard X-Ray Sources." The Astrophysical Journal Letters 941, no. 2 (2022): L22.
19. Grant Johnson, Patrick Kilian, Fan Guo, and **Xiaocan Li**. "Particle Acceleration in Magnetic Reconnection with Ad hoc Pitch-angle Scattering." The Astrophysical Journal, 933:73 (9pp), 2022 July 1
18. Oka, M., T. D. Phan, M. Øieroset, D. L. Turner, J. F. Drake, **X. Li**, S. A. Fuselier et al. "Electron energization and thermal to non-thermal energy partition during earth's magnetotail reconnection." Physics of Plasmas 29, no. 5 (2022): 052904.
17. Yi-Hsin Liu, Paul Cassak, **Xiaocan Li**, Michael Hesse, Shan-Chang Lin, and Kevin Genestreti. "First-principles theory of the rate of magnetic reconnection in magnetospheric and solar plasmas." Communications Physics 5, no. 1 (2022): 1-9.
16. Xiangliang Kong, Jing Ye, Bin Chen, Fan Guo, Chengcai Shen, **Xiaocan Li**, Sijie Yu, Yao Chen, and Joe Giacalone. "A model of double coronal hard X-ray sources in solar flares." The Astrophysical Journal,933:93 (8pp), 2022 July 1

15. Fan Guo, Lulu Zhao, Christina MS Cohen, Joe Giacalone, Richard A. Leske, Mark E. Wiedenbeck, Stephen W. Kahler, **Xiaocan Li**, Qile Zhang, George Ho, and Mihir Desai. "Variable Ion Compositions of Solar Energetic Particle Events in the Inner Heliosphere: A Field Line Braiding Model with Compound Injections." The Astrophysical Journal 924, no. 1 (2022): 22.
14. Qile Zhang, Fan Guo, William Daughton, **Xiaocan Li**, and Hui Li. "Efficient Nonthermal Ion and Electron Acceleration Enabled by the Flux-Rope Kink Instability in 3D Nonrelativistic Magnetic Reconnection." Physical Review Letters 127, no. 18 (2021): 185101.
13. Matthew Goodbred, Yi-Hsin Liu, Bin Chen, and **Xiaocan Li**. "The relation between the energy conversion rate and reconnection rate in Petschek-type reconnection—Implications for solar flares." Physics of Plasmas 28, no. 8 (2021): 082103.
12. Shan-Chang Lin, Yi-Hsin Liu, and **Xiaocan Li**. "Fast magnetic reconnection induced by resistivity gradients in 2D magnetohydrodynamics." Physics of Plasmas 28, no. 7 (2021): 072109.
11. Liping Yang, Hui Li, Fan Guo, **Xiaocan Li**, Shengtai Li, Jiansen He, Lei Zhang, and Xueshang Feng. "Fast Magnetic Reconnection with Turbulence in High Lundquist Number Limit." The Astrophysical Journal Letters 901, no. 2 (2020): L22.
10. Fan Guo, Yi-Hsin Liu, **Xiaocan Li**, Hui Li, William Daughton, and Patrick Kilian. "Recent progress on particle acceleration and reconnection physics during magnetic reconnection in the magnetically-dominated relativistic regime." Physics of Plasmas 27, no. 8 (2020): 080501.
9. Senbei Du, Gary P. Zank, **Xiaocan Li**, and Fan Guo. "Energy dissipation and entropy in collisionless plasma." Physical Review E 101, no. 3 (2020): 033208.
8. Yi-Hsin Liu, Shan-Chang Lin, Michael Hesse, Fan Guo, **Xiaocan Li**, Haocheng Zhang, and Sarah Peery. "The Critical Role of Collisionless Plasma Energization on the Structure of Relativistic Magnetic Reconnection." The Astrophysical Journal Letters, 892 L13
7. Fu, Xiangrong, Fan Guo, Hui Li, and **Xiaocan Li**. "Heating of Heavy Ions in Low-beta Compressible Turbulence." The Astrophysical Journal, 890(2), 161, 2020.
6. Chen, G., H. S. Fu, Y. Zhang, **Xiaocan Li**, Y. S. Ge, A. M. Du, C. M. Liu, and Y. Xu. "Energetic Electron Acceleration in Unconfined Reconnection Jets." The Astrophysical Journal Letters 881, no. 1 (2019): L8.
5. Adam Stanier, William Daughton, Ari Le, **Xiaocan Li**, and Robert Bird. "Influence of 3D plasmoid dynamics on the transition from collisional to kinetic reconnection." Physics of Plasmas 26, 072121 (2019)
4. Senbei Du, Fan Guo, Gary P. Zank, **Xiaocan Li**, and Adam Stanier. "Plasma Energization in Colliding Magnetic Flux Ropes." The Astrophysical Journal 867, no. 1 (2018): 16.
3. Xiangrong Fu, Hui Li, Fan Guo, **Xiaocan Li**, and Vadim Roytershteyn. "Parametric Decay Instability and Dissipation of Low-frequency Alfvén Waves in Low-beta Turbulent Plasmas." The Astrophysical Journal 855, no. 2 (2018): 139.
2. Fan Guo, Hui Li, William Daughton, **Xiaocan Li**, and Yi-Hsin Liu. "Particle acceleration during magnetic reconnection in a low-beta pair plasma." Physics of Plasmas 23, no. 5 (2016): 055708.
1. Lorin Arnold, Gang Li, **Xiaocan Li**, and Yihua Yan. "Observation of flux-tube crossings in the solar wind." The Astrophysical Journal 766, no. 1 (2013): 2.

Conference Proceedings

4. Bin Dong, Patrick Kilian, **Xiaocan Li**, Fan Guo, Suren Byna, and Kesheng Wu. 2019. "Terabyte-scale Particle Data Analysis: An ArrayUDF Case Study". In Proceedings of the 31st International Conference on Scientific and Statistical Database Management (SSDBM '19). ACM, New York, NY, USA, 202-205.
3. Senbei Du, Gary P. Zank, Fan Guo, **Xiaocan Li**, and Adam Stanier. "Particle Acceleration in Interacting Magnetic Flux Ropes." In Journal of Physics: Conference Series, vol. 1100, no. 1, p. 012009. IOP Publishing, 2018.
2. Kirit Makwana, Hui Li, Fan Guo, and **Xiaocan Li**. "Dissipation and particle energization in moderate to low beta turbulent plasma via PIC simulations." In Journal of Physics: Conference Series, vol. 837, no. 1, p. 012004. IOP Publishing, 2017.
1. Brahmananda Dasgupta, Gang Li, **Xiaocan Li**, Abhay Ram, Qiang Hu, Gang Li, Gary P. Zank, Xianzhi Ao, Olga Verkhoglyadova, and James H. Adams. "Particle transport and acceleration in a chaotic magnetic field: Implications for seed population to solar flare and CME." In AIP Conference Proceedings-American Institute of Physics, vol. 1500, no. 1, p. 56. 2012.

White Papers

5. Oka, M., …, **Li, X.** et al., 2023. "Particle acceleration in solar flares with imaging-spectroscopy in soft X-rays." *Whitepaper #302 in the Decadal Survey for Solar and Space Physics* (*Heliophysics*) *2024-2033*. Bulletin of the AAS, 55(3). https://doi.org/10.3847/25c2cfeb.c1b1eb07
4. Chen, B., …, **Li, X.** et al., 2023. "Quantifying Energy Release in Solar Flares and Solar Eruptive Events: New Frontiers with a Next-Generation Solar Radio Facility." *Whitepaper #060 in the Decadal Survey for Solar and Space Physics* (*Heliophysics*) *2024-2033*. Bulletin of the AAS, 55(3). https://doi.org/10.3847/25c2cfeb.aa2ad1d0
3. Gary, D., …, **Li, X.** et al., 2023. "Frequency Agile Solar Radiotelescope." *Whitepaper #123 in the Decadal Survey for Solar and Space Physics* (*Heliophysics*) *2024-2033*. Bulletin of the AAS, 55(3). https://doi.org/10.3847/25c2cfeb.7ecd0da5
2. Guo, F., …, **Li, X.** et al., 2023. "Advancing Theory and Modeling Efforts in Heliophysics." *Whitepaper #144 in the Decadal Survey for Solar and Space Physics* (*Heliophysics*) *2024-2033*. Bulletin of the AAS, 55(3). https://doi.org/10.3847/25c2cfeb.d8579f9b
1. Ji, H., …, **Li, X.** et al., 2023. "Major Scientific Challenges and Opportunities in Understanding Magnetic Reconnection and Related Explosive Phenomena in Heliophysics and Beyond." *Whitepaper #192 in the Decadal Survey for Solar and Space Physics* (*Heliophysics*) *2024-2033*. Bulletin of the AAS, 55(3). https://doi.org/10.3847/25c2cfeb.e22a8d1f

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Invited Talks

16. Understanding Particle Acceleration during Solar Flares through Integrated Modeling and Observations, *2023 AGU Fall Meeting*, Dec. 11th, 2023, San Francisco, CA.
15. Modeling Particle Acceleration and Transport in Solar Flares and Solar Eruption Regions, *EOVSA Data and GX Simulator Modeling Camp & FASR Workshop 2023*, Jan. 11th, 2023, Newark, NJ.
14. Modeling Electron Acceleration and Transport in Magnetic Reconnection *AOGS 19th Annual Meeting*, Aug. 1st, 2022, Online
13. Thermal and Nonthermal Partition in Nonrelativistic Magnetic Reconnection *Magnetic Reconnection Workshop 2022*, May. 19th, 2022, Monterey, CA.
12. The acceleration of charged particles in nonrelativistic magnetic reconnection *2021 AGU Fall Meeting*, Dec. 16th, 2021, Online
11. Formation of Power-law Electron Energy Spectra in 3D Low-β Magnetic Reconnection, *62nd Annual Meeting of APS DPP*, Nov. 9, 2020, Online
10. Power-Law Generation of Accelerated Particles in 3D Reconnection, *4th Asia-Pacific Conference on Plasma Physics*, Oct. 28, 2020, Zoom online
9. Large-scale particle acceleration during magnetic reconnection in solar flares. *Solar Physics Webinar of Global Reach—SolFER Colloquium*, Aug. 21, 2020, Webex online
8. The formation of power-law energy spectrum in low-β magnetic reconnection. *Magnetic Reconnection and Particle Acceleration in Solar Flares*, Mar. 18-20, 2019, Los Alamos, NM
7. The formation of power-law energy spectrum in 3D low-β magnetic reconnection. *The 18th Annual International Astrophysics Conference*, Feb. 18-22, 2019, Pasadena, CA
6. The Roles of Fluid Compression and Shear in Particle Energization during Magnetic Reconnection. *Particle Transport and Energization in Turbulent Plasmas*, April 24-27, 2018, Zhuhai, Guangdong, China.
5. The Roles of Fluid Compression and Shear in Particle Energization during Magnetic Reconnection. *The 17th Annual International Astrophysics Conference*, March 5-9, 2018, Santa Fe, NM.
4. The Role of Fluid Compression in Particle Energization during Magnetic Reconnection. *2017 AGU Fall Meeting*, New Orleans, LA.
3. The Role of Fluid Compression in Particle Energization during Magnetic Reconnection. *The 16th Annual International Astrophysics Conference*, March 6-10, 2017, Santa Fe, NM
2. Nonthermally Dominated Electron Acceleration during Magnetic Reconnection in a Low-beta Plasma. *Plasma Energization: Exchanges between Fluid and Kinetic Scales*, May 4-6, 2015, Los Alamos, NM.
1. Particle acceleration in reconnection: a view from electric current, *Meso-scale Plasma Dynamics and Energetic Particles: Applications to Laboratory, Space, and Astrophysical Plasmas*, June 30 – July 1, 2014, Los Alamos, NM.

Other Talks

26. Turbulence and Associated Particle Acceleration and Transport in 3D Magnetic Reconnection, *65th Annual Meeting of APS DPP*, Nov. 2nd, 2023, Virtual
25. Modeling particle acceleration and transport in 3D solar flare regions, *SDU Qingdao Forum on Energy Particles and Radio Radiation*, Oct. 26th, 2023, Virtual
24. Turbulence Properties in 3D Magnetic Reconnection: Implications for Particle Acceleration and Transport, *Friends-of-Magnetosphere (FoM) seminar at LASP*, Oct. 17th, 2023, Online
23. Modeling Particle Acceleration and Transport during Solar Flares, *Seminar at Space Science Center @ University of New Hampshire*, Sept. 27th, 2023
22. Modeling Electron Acceleration and Transport in 3D Solar Flare Regions, *HSR workshop at NJIT*, Aug. 29th, 2023
21. Modeling Particle Acceleration and Transport in Solar Flares, *Seminar at National Space Science Center*, May 5th, 2023, Online
20. Modeling Particle Acceleration and Transport in Solar Flare Magnetic Reconnection Region, *2022 AGU Fall Meeting*, Dec. 16th, 2022, Chicago, IL
19. A Model for Nonthermal Particle Acceleration in Relativistic Magnetic Reconnection, *64th Annual Meeting of APS DPP*, Oct. 18th, 2022, Online
18. Modeling Large-scale Electron Acceleration and Transport in Solar Flare Magnetic Reconnection Region, *2021 AGU Fall Meeting*, Dec. 14th, 2021, Online
17. Modeling Large-scale Electron Acceleration in Solar Flare Magnetic Reconnection Region, *IPMU Workshop on "Particle Acceleration in Solar Flares and the Plasma Universe – Deciphering its features under magnetic reconnection"*, Nov. 18th, 2021, Online
16. A model for nonthermal particle acceleration in magnetic reconnection, *63rd Annual Meeting of APS DPP*, Nov. 9th, 2021, Online
15. Turbulence properties relevant to particle acceleration and transport in 3D reconnection, *MMS Community Workshop*, Oct. 20th, 2021, Waterville Valley, NH
14. Modeling Large-scale Electron Acceleration Associated with Magnetic Reconnection, *SolFER Spring 2021 Meeing*, May 25th, 2021, Online
13. Power-law Index of Energy Spectrum in Magnetically Dominated Systems, *62nd Annual Meeting of APS DPP*, Nov. 11, 2020, Online
12. Power-law formation of nonthermal electrons in low-β reconnection. *MMS telecom*, July 28, 2020, online Webex meeting
11. Power-law formation in 3D low-β magnetic reconnection. *DRIVE Science Center Collaboration on Solar Flare Energy Release (SolFER), Group 3*, July 9, 2020, online Zoom meeting
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9. Particle acceleration in compressible reconnection layer. *60th Annual Meeting of the APS Division of Plasma Physics*, November 5–9, 2018; Portland, Oregon
8. Particle Acceleration during Magnetic Reconnection in Solar Flares. *Dartmouth College*, Oct. 23, 2018, Hanover, NH.
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